CKS-LEX-5-2026: Biology Domain Hierarchical Lexicons

Registry: [?]

Series Path: [CKS-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-13-2026] → [CKS-MATH-16-2026] → [CKS-DWDM-5-2026] → [CKS-MATH-17-2026] → [CKS-MATH-18-2026] → [CKS-MATH-19-2026] → [CKS-MATH-20-2026] → [CKS-MATH-21-2026]

Parent Framework: [CKS-0-2026]

DOI: 10.5281/zenodo.zzz

Date: February 2026

Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Date: March 1, 2026

Status: Biology-Specific Terminology Sets

Classification: Scalable Lexicons for Biology Papers

OPERATIONAL DECLARATION

This document provides CKS lexicons optimized for biology papers.

Purpose: Enable biologists to introduce CKS framework with appropriate depth.

Scope: Only terms relevant to biology applications:

- Disease mechanisms (cancer, heart, neurological)
- Cellular biology

- Organ systems
- Development and morphogenesis
- Healing and regeneration
- Metabolism and physiology
- Aging and death

Excluded: Pure physics, mathematics proofs, consciousness theory (see separate domain lexicons)

BIOLOGY DOMAIN REDUCTION SERIES

Level B0: Complete Biology Set (85 terms)

All biology-relevant terms from CKS framework

Foundational Axioms (6 terms)

- **D=3** - Three spatial dimensions
- **S=2** - Bilateral symmetry (body structure)
- ** ℚ ** - Rational substrate (discrete)
- **K-Space** - Discrete substrate (where biology computes)
- **X-Space** - Rendered perception (what organisms experience)
- **Discrete** - Countable, not continuous

Soliton Hierarchy (12 terms)

- **Soliton** - Stable geometric pattern (all living structures)
- **Tier** - Hierarchical nesting level
- **Tier-4** - Organism level (humans, animals, plants)
- **Tier-5** - Organ level (heart, liver, brain, kidney)
- **Tier-6** - Cell level (individual cells)
- **J-Distance** - Hierarchical depth (7.71 organism, 8.45 organ, 9.07 cell)
- **N=0 Pivot** - Universal root (all biology grounds here)
- **Registry** - Parent-child tracking chain
- **Registry Chain** - Path from cell → organ → organism → planet → N=0
- **Vortex** - Rotational soliton structure (organs, cells)

- **Toroidal** - Donut-shaped pattern (heart, organs)
- **Nested** - Hierarchical containment (cells in organs in organisms)

Health Metrics (15 terms)

- **R (Remainder)** - Accumulated processing residue
- **Δ = 19** - Healthy baseline R value
- **R = 69** - Malignancy threshold (critical)
- **θ (Angle)** - Registry pointer deviation (0°=health, 90°=disease)
- **$F(\theta) = \cos(\theta)$** - Flow efficiency function
- **Venting** - R-clearing mechanism (to parent soliton)
- **α (Alpha)** - Vent function (W=1, output/clearing)
- **β (Beta)** - Torque function (W=2, rotation/metabolism)
- **γ (Gamma)** - Socket function (W=3, containment/mass)
- **Φ_in = Φ_out** - Equilibrium condition (health)
- **dR/dt** - Remainder accumulation rate
- **Laminar Coherence** - Optimal state (R=19, θ=0°)
- **Turbulence** - Dysfunction state (R>19, θ>0°)
- **W-Functions** - Three geometric operations (α, β, γ)
- **Registry Pointer** - Direction vector to parent (angle θ)

Disease Mechanisms (18 terms)

- **90° Phase Lock** - Perpendicular blockage (venting failure)
- **6-9 Hook** - Self-sustaining closure at R=69 (cancer)
- **Toroidal Closure** - Donut R-accumulation (heart failure)
- **String-8 Closure** - Figure-8 fascial tension (cardiac arrest)
- **Side-B Inversion** - 180° flip (autoimmune)
- **Closure** - Topological venting blockage
- **Malignancy** - R=69 cellular closure (cancer)
- **Depression** - 60° angle (F=0.5, 50% reduction)
- **Anxiety** - 15-45° oscillation (instability)
- **Autism** - 90° hard metallic lock (bilateral failure)

- **Alzheimer's** - 90° soft viscous lock (temporal loop)
- **Psychosis** - 180° inversion (hallucinations)
- **Cancer** - 6-9 hook at cellular level
- **Cardiac Arrest** - Toroidal + string-8 combined
- **Heart Disease** - Toroidal closure in pericardium
- **Autoimmune** - Side-B visible (180°, self-attack)
- **Inflammation** - Partial R-accumulation ($\theta \approx 45°$)
- **Pathology** - Any closure or angular deviation

Biological Layers (8 terms)

- **Id-Layer** - Fascial/connective tissue (substrate interface)
- **Ib-Layer** - Local cellular/molecular processes
- **Fascia** - Connective tissue network (Id-layer primary)
- **ECM (Extracellular Matrix)** - Id-layer structure
- **Cellular** - Tier-6 level processes
- **Molecular** - Ib-layer chemistry
- **Tissue** - Id-layer organized structures
- **Systemic** - Tier-4/5 whole-organism level

Bilateral Structure (7 terms)

- **Bilateral** - Two-sided symmetry (S=2)
- **Side A / Side B** - Manifold faces (left/right)
- **Q-Operator** - A - B difference
- **Bilateral Reconciliation** - Left-right integration (15.19ms)
- **Hemispheric** - Brain bilateral structure
- **Symmetry** - S=2 reflection property
- **Mirror** - Bilateral reflection

Frequencies & Interventions (8 terms)

- **40 Hz** - Neural/gamma frequency (Alzheimer's treatment)
- **60 Hz** - Organ resonance (heart, liver, kidney)
- **32 Hz** - Base harmonic (biological rhythms)

- **110 Hz** - Information carrier frequency
- **Sonic Intervention** - Frequency therapy (40-60 Hz)
- **Resonance** - Matching natural frequency
- **Frequency** - Vibration rate (Hz)
- **Harmonic** - Integer multiple of base frequency

Measurement & Diagnostics (11 terms)

- **R-Proxy** - Biomarker correlating with R (HRV, inflammation)
- **HRV (Heart Rate Variability)** - Autonomic R-proxy (lower=higher R)
- **Inflammatory Markers** - CRP, IL-6, cortisol (R-proxies)
- **Metabolic Efficiency** - Glucose, mitochondrial function (R-inverse)
- **Bioimpedance** - Tissue resistance (R-correlation)
- **Angle Measurement** - Determining θ from function
- **Topological Diagnosis** - Identifying closure type
- **R-Density** - Remainder concentration in tissue
- **Closure Detection** - Imaging/identifying blockages
- **Functional Performance** - Output metrics (infer θ from F)
- **Flow Metrics** - Venting efficiency measurements

Treatment Protocols (10 terms)

- **R-Reduction Protocol** - Systematic lowering of R
- **Sleep** - Nightly R-clearing (8+ hours)
- **Fasting** - Reduced metabolic R-generation (16:8 or longer)
- **Exercise** - Kinetic R-venting
- **Postural Alignment** - Vertical correction ($\theta \to 0°$)
- **Registry Clearing** - R-flush to baseline
- **Topological Unwinding** - Reversing closures
- **Stress Reduction** - Lowering input R-rate
- **Coherence Maintenance** - Daily R=19 practices
- **Accept All/Deny None** - Behavioral algorithm (R=19)

Special Constants (5 terms)

- **15.19ms Lag** - Bilateral reconciliation time ($J \times S = 7.71 \times 2$)
- **1024-bit Sovereignty** - Maximum coherence threshold
- **Δ = 19** - Baseline (healthy remainder)
- **R = 69** - Disease threshold
- **J = 7.71, 8.45, 9.07** - Organism, organ, cell depths

Advanced Concepts (5 terms)

- **LERP** - K→X interpolation creating qualia
- **Qualia** - Subjective experience (bilateral integration)
- **Rendering** - K→X transform process
- **Death** - Registry decoupling from parent tier
- **Reincarnation** - Registry pattern persistence (theoretical)

Total: 85 terms

Level B1: Standard Biology Paper (50 terms)

Sufficient for most biology publications

Term	Biology Definition
D=3, S=2, $\mathbb{Q}$	Three axioms: 3D, bilateral symmetry, rational substrate
K-Space	Discrete substrate (computational biology domain)
Soliton	Stable pattern (cells, organs, organisms)
Tier-4, Tier-5, Tier-6	Organism, organ, cell levels
J-Distance	Hierarchical depth (7.71, 8.45, 9.07 for T4-6)
Registry	Parent-child tracking chain
N=0 Pivot	Universal root (all biology grounds here)
R (Remainder)	Accumulated processing residue
Δ = 19	Healthy baseline R
R = 69	Malignancy threshold
θ (Angle)	Registry deviation (0°=health, 90°=disease)
$F(\theta) = \cos(\theta)$	Flow efficiency ($F(0°)=1$, $F(90°)=0$)
Venting (α)	R-clearing to parent soliton
β, γ	Torque (metabolism), Socket (mass)

Term	Biology Definition
Φ_in = Φ_out	Health equilibrium
90° Phase Lock	Perpendicular venting blockage
6-9 Hook	Self-sustaining closure (R=69, cancer)
Toroidal Closure	Donut R-accumulation (heart)
String-8 Closure	Figure-8 fascial tension
Side-B Inversion	180° autoimmune mechanism
Depression (60°)	Partial topple, F=0.5
Autism (90° hard)	Metallic lock, bilateral failure
Alzheimer's (90° soft)	Viscous lock, temporal loop
Cancer	6-9 cellular closure at R=69
Cardiac Arrest	Toroidal + string-8 combined
Autoimmune	Side-B visible (180°, self-attack)
Id-Layer	Fascia/connective (substrate interface)
Ib-Layer	Cellular/molecular (local chemistry)
Bilateral (S=2)	Two-sided body structure
Q = A - B	Bilateral integration
40 Hz	Neural frequency (Alzheimer's treatment)
60 Hz	Organ resonance (therapeutic)
R-Proxy	Biomarker (HRV, inflammation, metabolic)
HRV	Heart rate variability (lower→higher R)
R-Reduction	Sleep, fast, exercise, stress↓
Postural Alignment	Vertical correction (θ→0°)
Sonic Intervention	40-60 Hz frequency therapy
Registry Clearing	R-flush to Δ=19
15.19ms Lag	Bilateral reconciliation (J × S)
1024-bit	Sovereignty threshold (R=19 required)
Laminar Coherence	Optimal R=19, θ=0° state
Closure	Topological venting blockage
Inflammation	Partial R-accumulation (θ≈45°)
Metabolic	β-function (torque/energy)
Tissue	Id-layer organized structure
Cell	Tier-6 soliton
Organ	Tier-5 soliton
Death	Registry decoupling from parent
Measurement	R-proxies, angle inference, function
Falsification	Specific testable predictions

Total: 50 terms

Level B2: Biology Methods Section (30 terms)

Core biological methodology

Term	Essential Biology Definition
D=3, S=2, $\mathbb{Q}$	Framework axioms
Soliton	Stable biological pattern
Tier-4/5/6	Organism/organ/cell
J-Distance	7.71, 8.45, 9.07
Registry	Parent-child chain
R (Remainder)	Processing residue
Δ=19 / R=69	Health / Disease thresholds
θ (Angle)	0°=health, 90°=disease
F(θ)=cos(θ)	Flow efficiency
Venting (α)	R-clearing mechanism
90° Lock	Perpendicular blockage
6-9 Hook	Cancer closure (R=69)
Toroidal	Heart failure pattern
Depression (60°)	F=0.5 reduction
Autism (90°)	Hard lock
Alzheimer's (90°)	Soft lock
Cancer	Cellular 6-9 hook
Autoimmune	180° Side-B visible
Id-Layer	Fascia/connective
Ib-Layer	Cellular/molecular
40Hz/60Hz	Neural/organ frequencies
R-Proxy	HRV, inflammation, metabolic
R-Reduction	Sleep, fast, exercise
Postural	Vertical alignment
Sonic	Frequency therapy
15.19ms	Bilateral lag
Bilateral (S=2)	Two-sided structure
Closure	Venting blockage
Measurement	Biomarkers, function
Falsification	Testable predictions

Total: 30 terms

Level B3: Biology Abstract (20 terms)

Minimal biological context

Term	Abstract-Level Definition
D=3, S=2, $\mathbb{Q}$	Geometric axioms
Soliton	Biological pattern
Tier-4/5/6	Organism/organ/cell
R (Remainder)	Health metric
Δ=19 / R=69	Health/disease
θ (Angle)	0°=health, 90°=disease
F(θ)=cos(θ)	Efficiency
90° Lock	Venting blockage
6-9 Hook	Cancer (R=69)
Depression (60°)	Partial topple
Autism (90°)	Hard lock
Cancer	Cellular closure
Id-Layer	Fascia
40Hz/60Hz	Therapeutic frequencies
R-Proxy	Biomarkers
R-Reduction	Treatment protocol
Bilateral	S=2 symmetry
Measurement	Validation
Falsification	Tests
Registry	Hierarchy

Total: 20 terms

Level B4: Biology Ultra-Compact (15 terms)

Minimum for comprehension

Term	Ultra-Compact
D=3, S=2, $\mathbb{Q}$	Axioms
Soliton	Pattern
Tier	Level
R	Remainder
Δ=19/R=69	Health/disease
θ	Angle
90° Lock	Blockage
6-9 Hook	Cancer
Depression	60°
Autism	90° hard
40Hz/60Hz	Frequencies
R-Proxy	Biomarker
R-Reduction	Treatment

Term	Ultra-Compact
Measurement	Validation
Bilateral	S=2

Total: 15 terms

Level B5: Biology Core (10 terms)

Skeleton understanding

Term	Core Biology
D=3, S=2, $\mathbb{Q}$	Axioms
Soliton	Pattern (cell/organ)
R	Remainder
Δ=19/R=69	Health/disease
θ	0°/90°
90° Lock	Disease mechanism
Cancer	R=69
40Hz/60Hz	Treatment
R-Proxy	Measurement
Bilateral	S=2

Total: 10 terms

Level B6: Biology Absolute Minimum (7 terms)

Cannot reduce further

Term	Irreducible Biology
D=3, S=2, $\mathbb{Q}$	Framework axioms (3D, bilateral, rational)
Soliton	Biological pattern (cell, organ, organism)
R (Remainder)	Health metric (Δ=19 health, R=69 disease)
θ (Angle)	Registry deviation (0° optimal, 90° pathological)
90° Lock	Disease mechanism (venting blockage)
R-Proxy	Biomarker (HRV, inflammation, metabolic)

Term	Irreducible Biology
Treatment	R-reduction, frequency, postural alignment

Total: 7 terms

Level B7: Emergency Biology (5 terms)

Severe constraint only

Term	Emergency
D=3, S=2, $\mathbb{Q}$	Geometric axioms
R	Remainder (health metric)
θ	Angle (0°=health, 90°=disease)
Disease	R=69, 90° closures
Treatment	R-reduction protocols

Total: 5 terms

Level B8: Biology Minimum (3 terms)

Single sentence only

Term	Absolute Minimum
R, θ	Health metrics
Disease	Topological closure
Treatment	Geometric correction

Total: 3 terms

BIOLOGY-SPECIFIC USAGE GUIDE

By Biology Sub-Domain

Cancer Biology Papers

Priority additions to base level:

- R=69 threshold, 6-9 hook
- Tier-6 (cellular), malignancy
- α-vent blockage, β-γ closure
- Metabolic independence
- R-density measurement
- Frequency intervention (60 Hz)
- Topological closure detection

Cardiovascular Papers

Priority additions:

- Toroidal closure, string-8
- Tier-5 (organ), heart vortex
- Pericardial fluid, fascia
- Postural correlation
- 60 Hz intervention
- HRV as R-proxy
- Cardiac arrest mechanism

Neuroscience Papers

Priority additions:

- 40 Hz gamma, neural frequencies
- Bilateral hemispheric structure
- 15.19ms lag, LERP
- Alzheimer's (90° soft), autism (90° hard)
- Depression (60°), anxiety (oscillation)
- Temporal loops, spatial locks
- Side-B inversion (psychosis)

Cellular Biology

Priority additions:

- Tier-6 soliton
- J=9.07 (cell depth)

- Ib-layer (molecular)
- Id-layer interface
- Registry chain cell→organ→organism
- Cellular metabolism (β)
- Apoptosis vs 6-9 hook

Physiology/Metabolism

Priority additions:

- β-function (torque/metabolism)
- Φ_in = Φ_out (equilibrium)
- dR/dt (accumulation rate)
- Sleep (R-clearing)
- Fasting (R-reduction)
- Exercise (kinetic venting)
- Metabolic efficiency as R-proxy

Immunology

Priority additions:

- Autoimmune (180° inversion)
- Side-B visible (self-attack)
- Bilateral failure
- Inflammation (45° partial)
- Immune response as correct (topology wrong)
- Treatment: restore θ<90°

Development/Morphogenesis

Priority additions:

- Tier progression (cell→organ→organism)
- Registry chain formation
- Bilateral development (S=2)
- Id-layer structuring (fascia)
- J-distance establishment

- Geometric template

Common Biology Paper Structures

Research Article

Abstract: Level B3 (20 terms)

Introduction: Level B2 (30 terms)

Methods: Level B1 (50 terms)

Results: Level B2 (30 terms)

Discussion: Level B1 (50 terms)

Conclusion: Level B3 (20 terms)

Case Study

Abstract: Level B4 (15 terms)

Background: Level B3 (20 terms)

Case: Level B2 (30 terms)

Discussion: Level B3 (20 terms)

Review Article

Abstract: Level B3 (20 terms)

Introduction: Level B2 (30 terms)

Framework: Level B1 (50 terms)

Disease mechanisms: Level B0 (85 terms)

Treatment: Level B1 (50 terms)

Future: Level B2 (30 terms)

Biology Term Introduction Order

Optimal sequence for biology papers:

1. **Axioms** (D=3, S=2, $\mathbb{Q}$) - "Biology operates on discrete substrate…"
2. **Soliton** - "Cells, organs, organisms are stable patterns…"
3. **Tiers** - "Hierarchical: cell (T6) in organ (T5) in organism (T4)…"

4. **R (Remainder)** - "Accumulated residue, health metric…"
5. **θ (Angle)** - "Vertical alignment, 0°=health, 90°=disease…"
6. **F(θ)** - "Flow efficiency drops with angle…"
7. **Disease mechanisms** - "90° locks, 6-9 hooks, closures…"
8. **Treatment** - "R-reduction, frequency, postural…"
9. Then sub-domain specific terms

Critical Biology Equations

Always include when relevant:

Health Metrics

$F(\theta) = F\_max \times \cos(\theta)$

- $F(0°) = 1.0$ (optimal)
- $F(60°) = 0.5$ (depression)
- $F(90°) = 0$ (complete blockage)

$dR/dt = \Phi\_in - \Phi\_out$

- Health: $dR/dt = 0$ ($\Phi\_in = \Phi\_out$, $R = \Delta = 19$)
- Disease: $dR/dt > 0$ ($R \to 69$)

Jacobian Partition (R=69)

At malignancy:

γ (socket) = 49.3 ($5/7 \times 69$)

β (torque) = 19.7 ($2/7 \times 69$)

α (vent) = 0 (blocked)

Total = 69

Tier Depths

J-distance (logarithmic units):

Tier-4 (organism): J = 7.71

Tier-5 (organ): J = 8.45

Tier-6 (cell): J = 9.07

Bilateral lag: $J \times S = 7.71 \times 2 = 15.42 \approx 15.19$ms

R-Proxy Correlations

HRV (Heart Rate Variability): Lower → Higher R

Inflammation (CRP, IL-6): Higher → Higher R

Metabolic (glucose, mito): Lower → Higher R

Coherence (EEG, HRV): Higher → Lower R

Biology-Specific Falsification

Tests specific to biology domain:

1. **R=69 threshold**: Measure R-density in healthy vs cancerous tissue
2. **90° phase lock**: Image neural/cardiac flow, calculate angles
3. **40 Hz efficacy**: Alzheimer's patients, double-blind 40 Hz vs placebo
4. **60 Hz cardiac**: Heart failure patients, measure response to 60 Hz
5. **Postural correlation**: Spinal alignment vs disease progression
6. **HRV as R-proxy**: HRV vs inflammatory markers vs disease states
7. **Bilateral lag**: 15.19ms proprioceptive delay measurement
8. **Closure detection**: Imaging modalities show predicted topologies

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Comparison to Standard Biology

Key distinctions to emphasize:

Standard Biology	CKS Framework
Chemical mechanisms	Topological mechanisms
Genetic mutations cause cancer	R=69 closure causes cancer
Plaque causes heart disease	Toroidal+string-8 closure
Amyloid causes Alzheimer's	90° temporal lock
Random disease onset	Geometric threshold (R=69, θ=90°)
Symptom treatment	Topological correction
Many disease types	Few geometric patterns
Each disease unique	Universal closure mechanisms

Disease Classification by Geometry

By Angle (θ)

0-15°: Optimal health

15-45°: Mild dysfunction (anxiety, mild inflammation)

45-60°: Moderate dysfunction (chronic conditions)

60°: Depression (F=0.5, 50% reduction)

90°: Critical (autism hard, Alzheimer's soft, cancer)

180°: Inversion (autoimmune, psychosis)

By Closure Type

6-9 Hook (R=69): Cancer, addiction, obsession

Toroidal: Heart failure, organ congestion

String-8: Cardiac arrest, fascial restriction

Side-B Inversion: Autoimmune diseases

Bilateral failure: Autism, some psychoses

Temporal loop: Alzheimer's, dementia

By Tier

Tier-4 (Organism): Depression, anxiety, systemic

Tier-5 (Organ): Heart disease, cancer (organ-level)

Tier-6 (Cell): Cancer (cellular), cellular dysfunction

Treatment Protocol Templates

R-Reduction Protocol (Standard)

Daily:

- 8+ hours quality sleep (nightly R-flush)
- 16:8 eating window (reduced input)
- 30+ min exercise (kinetic venting)
- Vertical posture awareness (θ→0°)
- 10 min 60 Hz humming/toning (sonic)
- Stress management (lower input rate)

Weekly:

- 24-48 hour fast (deep R-reduction)
- Postural alignment session (structural)
- Assessment (R-proxies: HRV, markers)

Goal: $R \rightarrow \Delta = 19$ over 4-12 weeks

Cancer-Specific (Theoretical, Untested)

Goal: Break 6-9 hook, reduce $R < 69$

Protocol:

- Fasting 48-72 hrs pre/post intervention
- 60 Hz targeted sonic (tumor location)
- Postural alignment (open α-vent)
- Sleep optimization (maximum clearing)
- Stress elimination (minimize input)
- HRV/metabolic monitoring

Integration: WITH standard oncology care

Timeline: Pilot studies needed

Status: THEORETICAL - requires validation

Alzheimer's (40 Hz Protocol)

Based on existing research:

- 40 Hz light exposure (1 hour daily)
- 40 Hz auditory (1 hour daily)
- Music from youth (temporal anchoring)
- Vertical posture ($\theta \rightarrow 0°$)
- Sleep optimization (8+ hours)
- Low inflammation diet (R-reduction)

Evidence: Some published studies show benefit

CKS explanation: Disrupts 90° temporal toroid

Status: Partially validated, needs more research

Depression (Angle Correction)

Goal: Restore θ from 60° to 0°

Protocol:

- Postural: Vertical alignment awareness
- Exercise: Daily movement (kinetic venting)
- Sleep: 8-9 hours (R-clearing)
- Social: Connection (shared venting)
- Expression: Journaling, art, talking
- Sunlight: Circadian alignment
- 60 Hz: Daily humming/toning

Mechanism: Reduce R, straighten θ

Timeline: 4-8 weeks for noticeable change

Integration: Can combine with therapy, medication

R-Proxy Measurement Guide

Primary R-Proxies

HRV (Heart Rate Variability):

- Measure: SDNN, RMSSD
- Interpretation: Lower = Higher R
- Equipment: Heart rate monitor, HRV app
- Frequency: Daily to weekly

Inflammatory Markers:

- Measure: CRP, IL-6, cortisol
- Interpretation: Higher = Higher R
- Equipment: Blood test
- Frequency: Monthly to quarterly

Metabolic:

- Measure: Glucose, HbA1c, mitochondrial function
- Interpretation: Worse = Higher R
- Equipment: Blood test, metabolic panel

- Frequency: Quarterly

Secondary R-Proxies

Sleep Quality:

- Measure: Sleep tracking, wake frequency
- Interpretation: Worse = Higher R

Physical Performance:

- Measure: Endurance, strength, recovery
- Interpretation: Worse = Higher R

Cognitive:

- Measure: Clarity, focus, memory
- Interpretation: Worse = Higher R

Subjective:

- Measure: “Heaviness,” brain fog, vitality
- Interpretation: Worse = Higher R

Composite R-Score

Estimate R from multiple proxies:

R_estimated = 19 + Σ(proxy deviations)

Example:

HRV: 20% below baseline = +10

CRP: 2 $\times$ elevated = +15

Sleep: Poor quality = +10

Fatigue: Moderate = +10

Estimated R $\approx$ 19 + 45 = 64 (approaching critical)

Action: Intensive R-reduction protocol

Recommended Biology Paper Template

Title

Use Level B8 (3 terms): “Disease as Topological Closure: R,θ Metrics in [Condition]”

Abstract (250 words)

Use Level B3 (20 terms):

- State condition/disease
- Introduce CKS framework (axioms)
- Present geometric mechanism
- Describe measurements
- State falsification tests

Introduction (1000 words)

Use Level B2 (30 terms):

- Disease background
- Current understanding limitations
- CKS axioms introduction
- Soliton hierarchy (tiers)
- R and θ as health metrics
- Preview of mechanism

CKS Framework (1500 words)

Use Level B1 (50 terms):

- Full axiom explanation
- Tier structure (4, 5, 6)
- R and θ definitions
- Flow efficiency F(θ)
- Closure types
- Bilateral structure

Disease Mechanism (2000 words)

Use Level B0 (85 terms):

- Specific closure type
- R-accumulation pathway
- Angular deviation process
- Why $\theta \to 90°$ or $R \to 69$

- Tier-specific manifestation
- Id vs Ib layer involvement

Measurements (1500 words)

Use Level B1 (50 terms):

- R-proxy selection
- Measurement protocols
- Angle inference methods
- Baseline vs diseased
- Statistical analysis
- Results

Treatment Protocol (1000 words)

Use Level B2 (30 terms):

- R-reduction approach
- Frequency intervention (if applicable)
- Postural correction (if applicable)
- Monitoring plan
- Expected timeline
- Safety considerations

Discussion (1500 words)

Use Level B1 (50 terms):

- Comparison to standard model
- Advantages of geometric view
- Limitations
- Integration with existing care
- Future research directions

Conclusion (500 words)

Use Level B3 (20 terms):

- Geometric mechanism summary

- Falsification tests specified
- Clinical implications
- Research needs

Critical Disclaimers for Biology Papers

ALWAYS include when discussing disease/treatment:

CRITICAL DISCLAIMERS:

1. THEORETICAL FRAMEWORK
    - CKS is mathematical model, not proven medical fact
    - Requires extensive empirical validation
    - Predictions are testable but not yet tested
2. NOT MEDICAL ADVICE
    - Do not replace evidence-based medical care
    - Consult healthcare providers for treatment
    - Standard care should continue
3. RESEARCH CONTEXT ONLY
    - Protocols described for research purposes
    - Not approved for clinical use
    - Require IRB approval and proper trials
4. FALSIFIABLE PREDICTIONS
    - Framework specifies exact tests
    - Could be proven wrong by measurements
    - Open to revision based on data
5. INTEGRATION APPROACH
    - Complementary to existing medicine
    - Not replacement or contradiction
    - Adds topological perspective

END CKS-LEX-5-2026

Status: Complete Biology Domain Lexicons

Levels: 8 (from 85 terms to 3)

Optimization: Biology papers only

Minimum: 7 terms (Level B6)

For other domains, see:

- CKS-LEX-3-2026: Physics Domain ✓
- CKS-LEX-4-2026: Mathematics Domain ✓
- CKS-LEX-6-2026: Consciousness Domain (next)
- CKS-LEX-7-2026: Engineering Domain
- CKS-LEX-8-2026: Clinical/Medical Domain

Biology lexicon complete with critical safety disclaimers.

IMPORTANT: All biological/medical applications require:

- Explicit theoretical status
- No medical advice claims
- Integration with standard care
- IRB approval for research
- Proper clinical trials before application

Biology lexicon complete and optimized with appropriate cautions.

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-13-2026**] The Resonant Epoch. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-13-2026

[**CKS-MATH-16-2026**] The Geometric Necessity of 32. Github: https://github.com/ghowland/cks/tree/main/papers/MAT
MATH-16-2026

[**CKS-DWDM-5-2026**] The Geometry of the 66th Harmonic. Github: https://github.com/ghowland/cks/tree/main/papers/I
DWDM-5-2026

[**CKS-MATH-17-2026**] The 7-Bubble Jacobian. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-17-2026

[**CKS-MATH-18-2026**] The 84-Bit Word on a 32-Bit Computer. Github: https://github.com/ghowland/cks/tree/main/paper
MATH-18-2026

[**CKS-MATH-19-2026**] Topological Recirculation. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CK
MATH-19-2026

[**CKS-MATH-20-2026**] The Winding Torus. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-20-2026

[**CKS-MATH-21-2026**] The Final Constant Closure. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
MATH-21-2026